

DNS — Guided Notes

Unit 1.6.2 | CompTIA Network+ N10-009 Obj. 1.6 | N10-008 1.6

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Match each DNS record type to its purpose:

A → _____

AAAA → _____

CNAME → _____

MX → _____

PTR → _____

Question 2

A TXT record stores _____ data in DNS. Common uses include _____ (proves domain ownership to email providers) and _____ (specifies which servers are authorized to send email for the domain).

Question 3

A forward lookup zone resolves _____ to _____. A reverse lookup zone resolves _____ to _____. Reverse lookups are used primarily for _____ and _____.

Question 4

An authoritative DNS server holds the _____ records for a zone and gives definitive answers. A recursive resolver queries _____ on behalf of clients, caching results for up to the record's _____ value.

Question 5

DNS cache poisoning occurs when _____. An attacker who succeeds can redirect users to _____. The defense against this attack is _____, which adds cryptographic signatures to DNS responses.

Question 6

Split-horizon DNS serves _____ DNS responses to internal clients and _____ DNS responses to external clients for the same domain name. This allows internal users to reach _____ while external users reach _____.

Question 7

The command _____ queries DNS and shows the record type, answer, and which server responded. The command _____ is a more detailed alternative available on Linux/Mac. To test a specific DNS server directly, the syntax is _____.

Question 8 — Real World Application

REAL WORLD: A user can reach internal file servers by IP but not by name. External websites resolve fine. nslookup on their workstation returns correct results for external domains but times out for internal hostnames. What is the most likely cause and how do you fix it?

Download the answer key from the class server:

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wget  
http://[SERVER]/resources/network-engineer/1-6-2-dns/1-6-2-dns-answer-key.pdf
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